Guidodyno Software Setup

1 Overview

The Guidodyno comes with comprehensive software to make data collection and visualization easy. The user software stack ingests data from the DAQ module, stores it in a database (InfluxDB), and displays it using Grafana. This guide contains step by step instructions for how to configure this system on a Windows 11 computer.

At this time, the system does not virtualize. Keep in mind these steps will install the software on the top level on the computer, potentially interfering with other programs.

2 Installation

2.1 System Requirements

- Windows 11
- 8 GB RAM
- Ethernet Network Adapter [1]

2.2 Dependencies

2.2.1 Python

Python 3.13 or later must be installed. To check this, open command prompt, and run the following:

```
python --version
```

You should see:

```
Python 3.x.x
```

If you do not see your python version, or you do not have Python 3.13+, install the latest version of python (here) on your computer before proceeding.

Long PATH Names

Depending on your computer, the python installer may ask to adjust path lengths, universal path settings, or install CPython. Respond "y" to these requests.

2.2.2 Github Desktop

Make sure Github desktop is installed on your computer.[2] Default settings are OK. Then sign in to your Github account.

Download the latest version here

[1] Even if your computer does not have a physical Ethernet port, you likely still have an virtual Ethernet network adapter. You will still need an USB to Ethernet adapter dongle

[2] Install the Github desktop application even if you already have Git, unless you are sophisticated enough to set up Git LFS on your own

2.2.3 Grafana

Download Grafana from this website. Make sure to download it here, instead of the Grafana Cloud website. Select version 12.3.2 Enterprise edition.

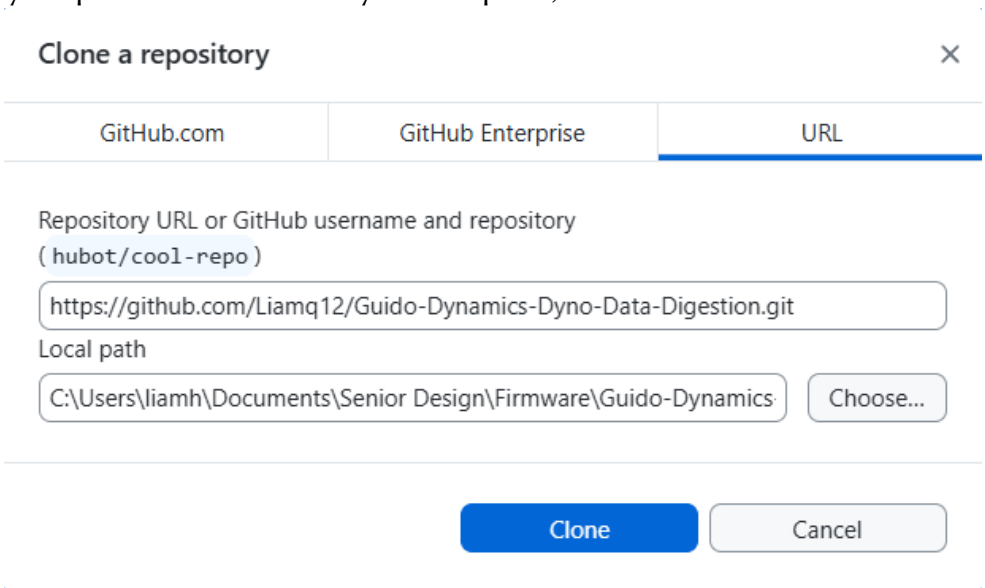
Keep all default settings while going through the Grafana setup steps. [3]

2.3 Repository Setup

Open Github desktop and navigate:
File → Clone Repository Select the URL menu, and enter the following URL: [4]

```
https://github.com/Liamq12/Guido-Dynamics-Dyno-Data-Digestion.git
```

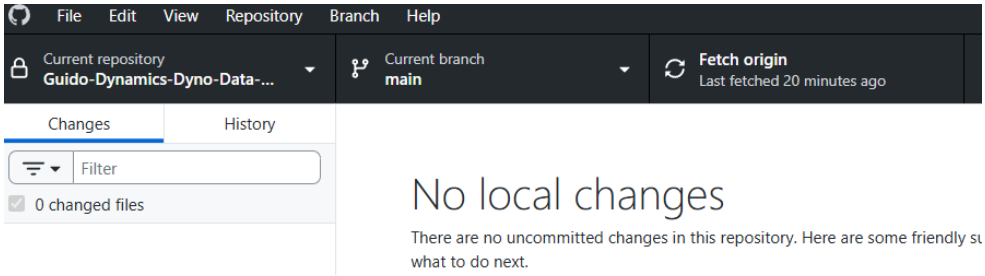
Select a path to your preferred location on your computer, and click "Clone".



If prompted, click "Initialize Git LFS". Once the repository has cloned, click "Fetch Orgin" a the top of the Github desktop application. Make sure to fetch orgin periodically to get the latest software updates.

[3] The default installation directory should be called "GrafanaLabs" in your Program Files directory

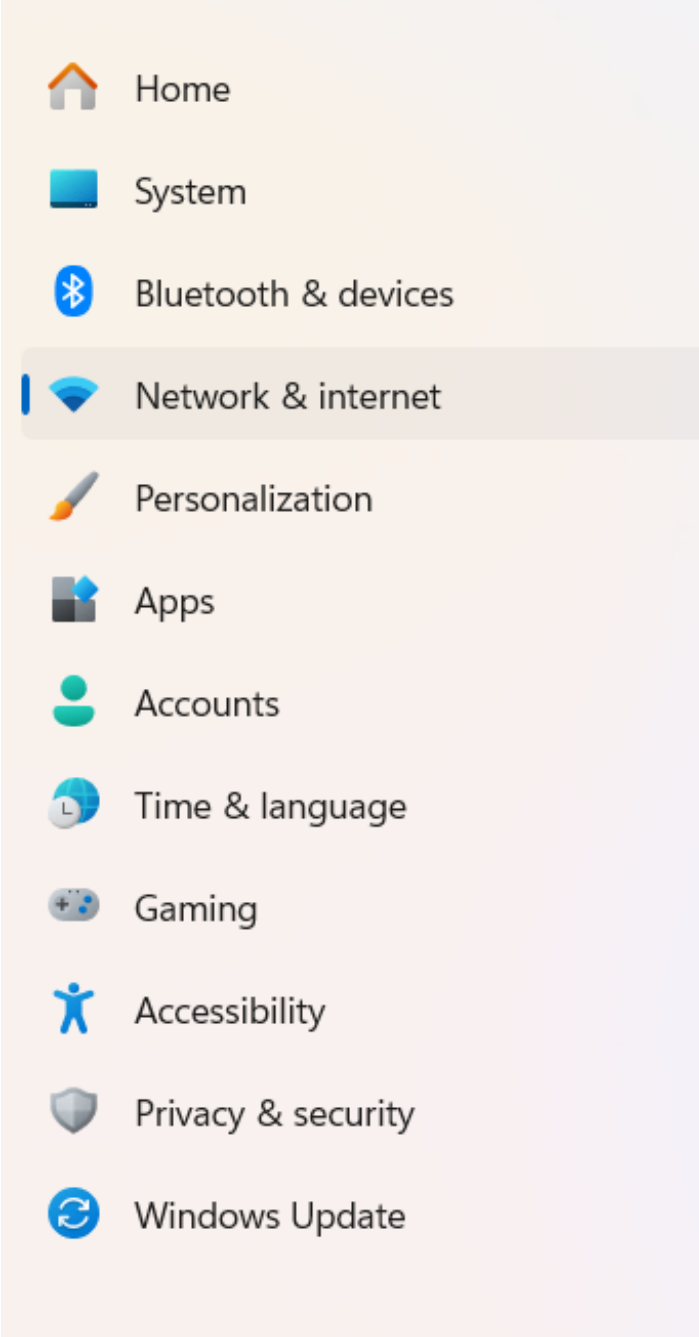
[4] If you are installing this after June 2026, the repository should be public, and should allow you to clone. If before June 2026, you may need to request access to the repository



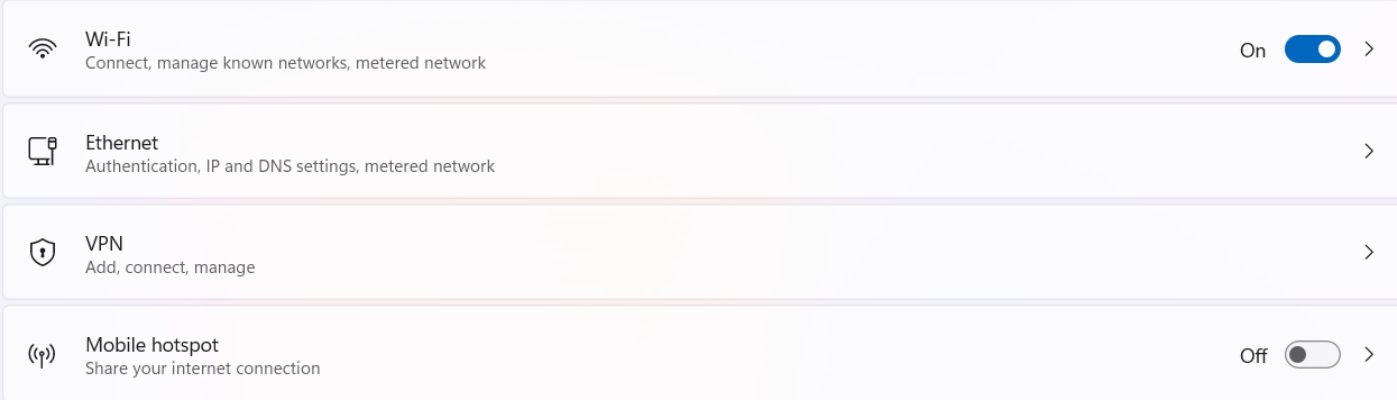
The Guidodyno software should now be installed in the directory you listed from the cloning step.

2.4 Network Settings

In the windows start menu, search for "Settings" and navigate to the "Network and Internet" tab.



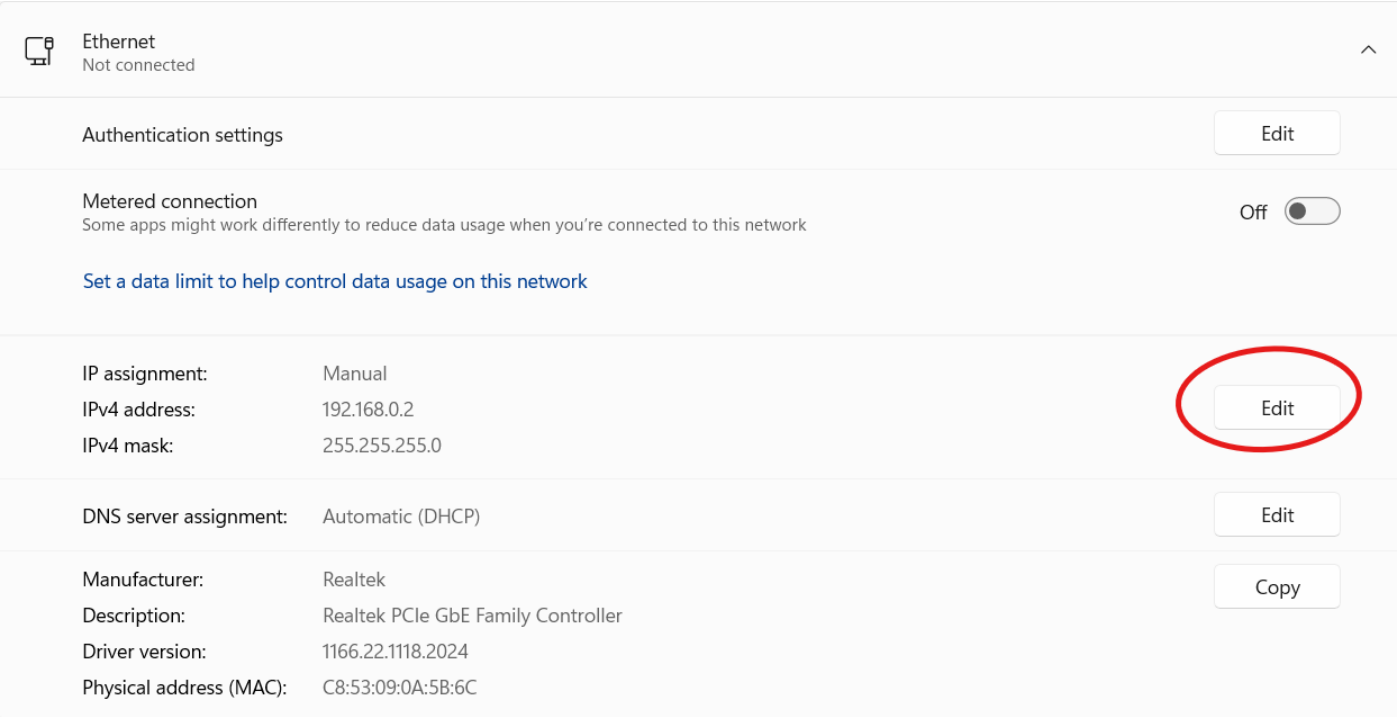
If you have a physical Ethernet port, click the menu button called "Ethernet"



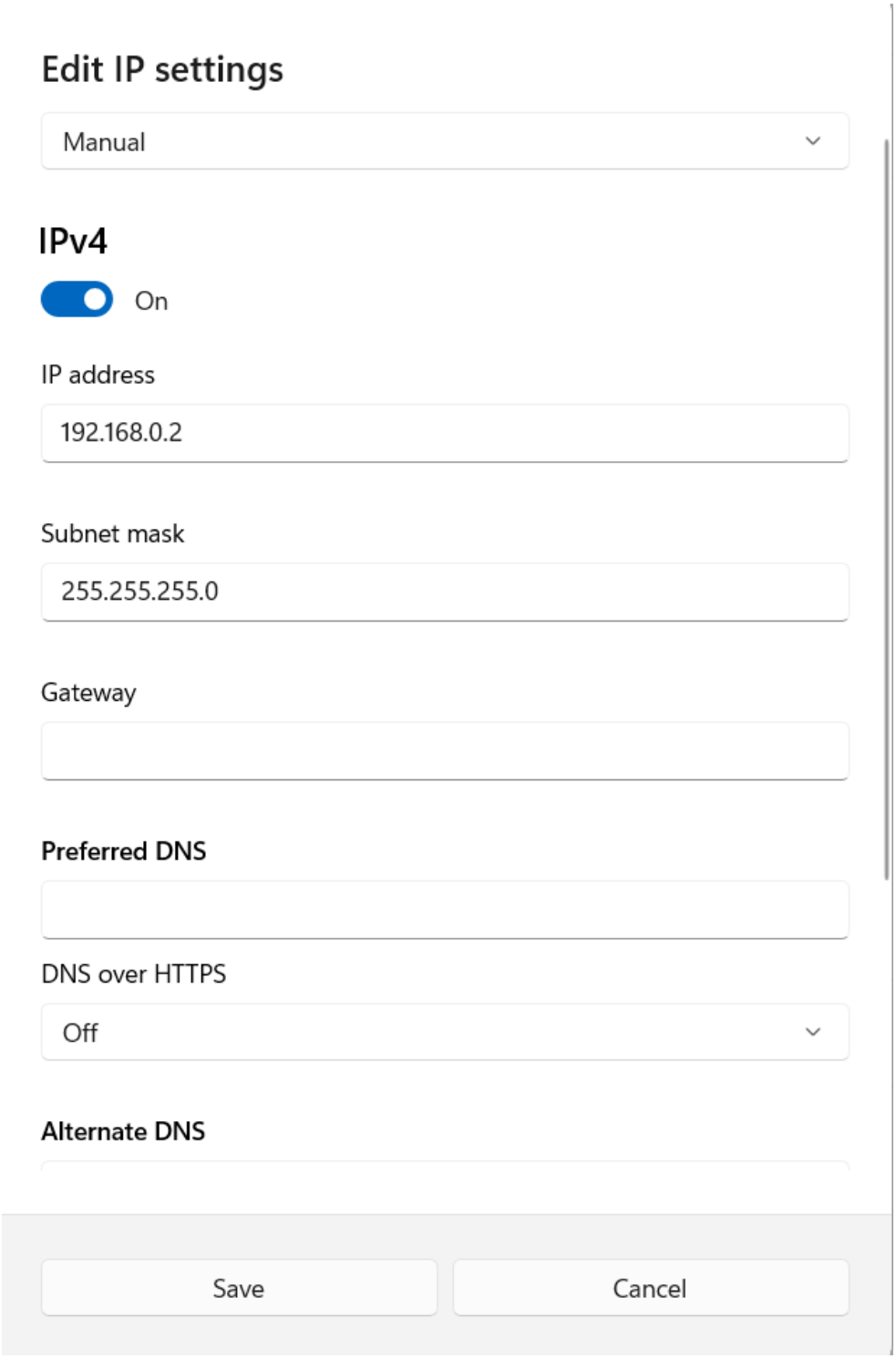
No Ethernet Menu

If you do not have an physical Ethernet port on your computer, you may need to navigate further menus to find this Ethernet network settings menu.

Click the second "Edit" button, circled below.



Under "Edit IP Settings", select manual.
Turn on IPv4.
For IP address, enter "192.168.0.2".
For subnet mask, enter "255.255.255.0".
Leave gateway field empty.
Leave perferred DNS field empty.
Under "DNS over HTTPS", select off.
Leave alternate DNS field empty.
Again, under "DNS over HTTPS", select off.
Turn off IPv6.
Click save.



2.5 runMe.bat

The runMe.bat file, located in the top level of your cloned repository, will start the necessary software to operate Guidodyno. Navigate to this file, and run as administrator. [5]

This script starts the necessary software (InfluxDB, Grafana, UDPtoInflux Bridge "main.py", and the user terminal) and will spawn many windows. To close these windows navigate to the first CMD window, and press any key to quit. This will close all spawned windows, and shutdown services correctly.

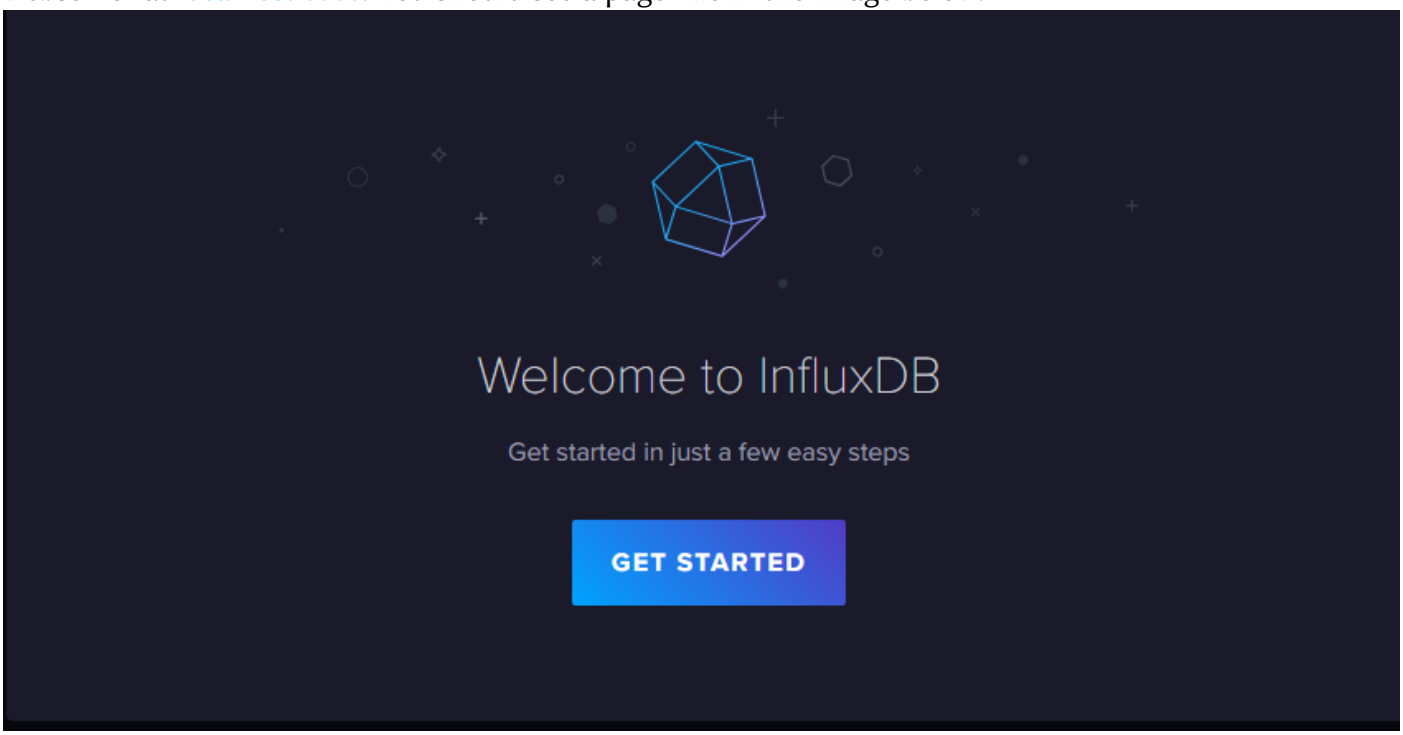
```
Starting UDP Ingest...
Starting User Terminal...
All processes started. Press any key to stop them...
Press any key to continue . . .
```

Program Stuck

The first time running this script often requires kicking the terminal. If the window is stuck at "The Grafana service was stopped successfully", press the enter key once to restart the program.

2.6 Configuring InfluxDB

With the script running from the previous step, open a new browser tab and go to the InfluxDB local webserver at localhost:8086. You should see a page like in the image below.

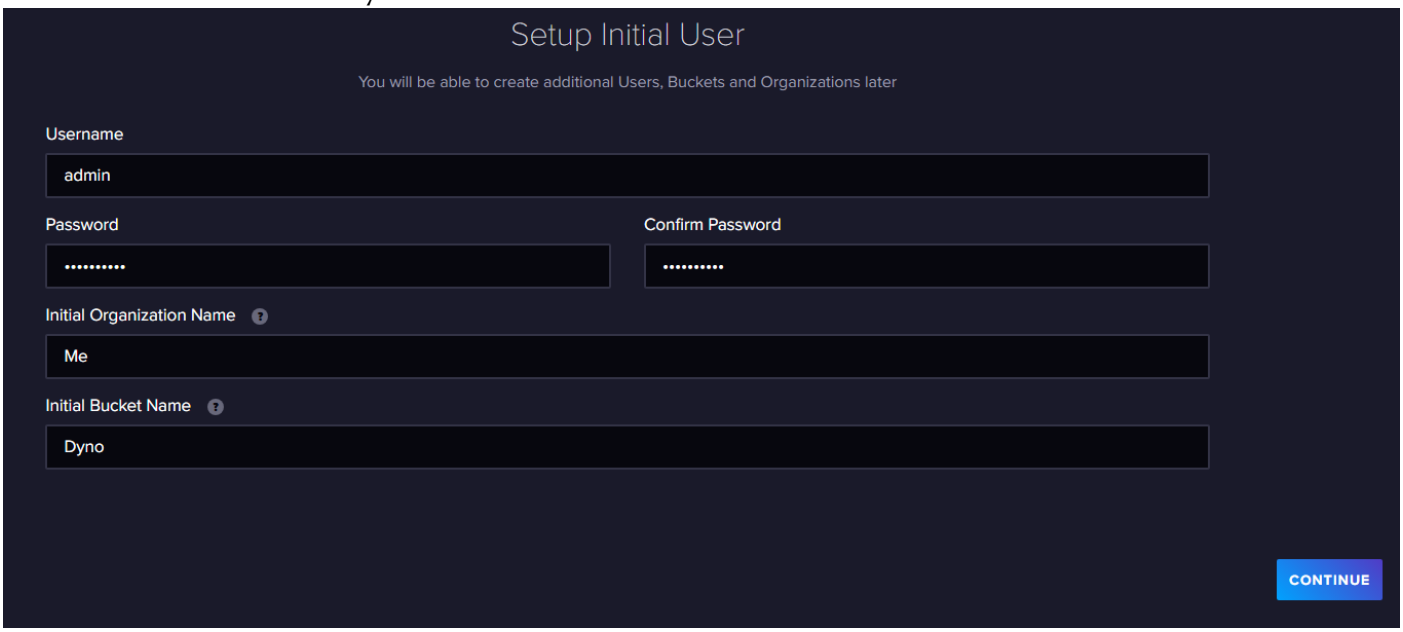


Error: Localhost Refused to Connect

Ensure the runMe.bat program is running (as administrator), and that you have fetched the latest version of the repository (With Git LFS enabled) from Github Desktop.

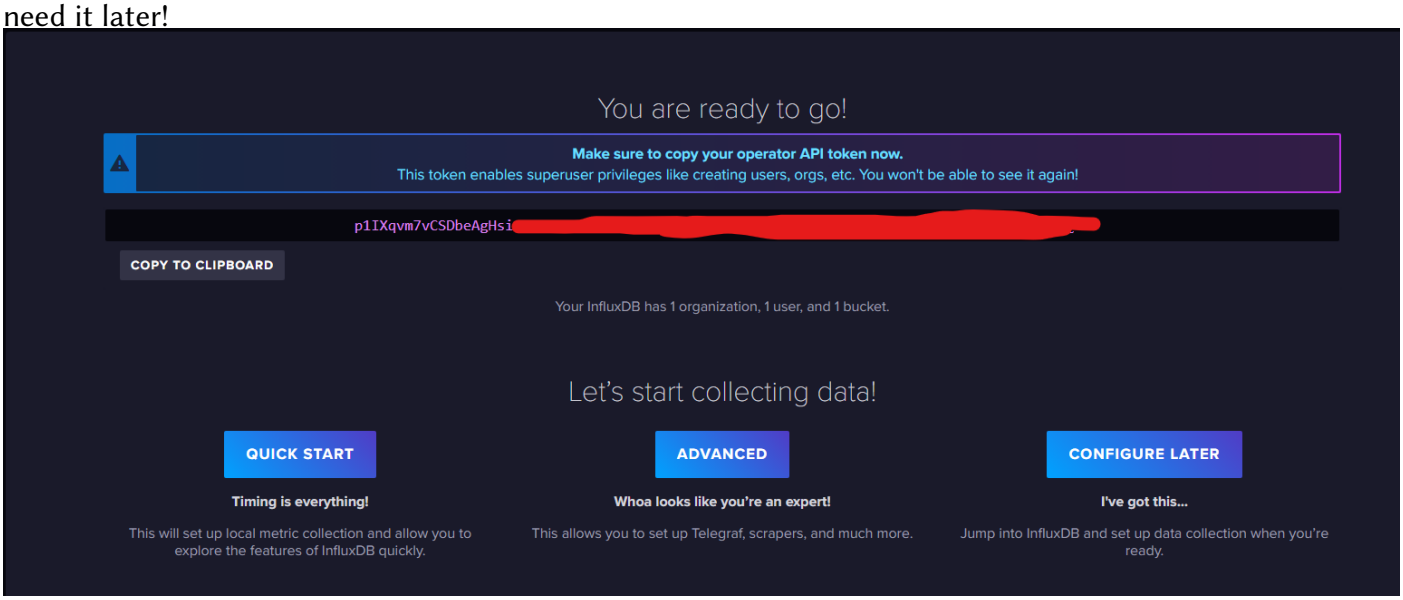
[5] This file is located with your other files cloned from GitHub. Unless changed in the clone step, the default location is: `C:\Users\[YOUR_USERNAME]\Documents\GitHub`

Click "Get Started" and make a username and password. [6] Set "Initial Organization Name" to "Me", and "Initial Bucket Name" to "Dyno". Click "Continue".



Since this information is stored locally, it is highly recommended to allow your browser to save your username and password.

You will now see your unique superuser token. This is only shown once, and you will not be able to view it again after moving from this page. Copy the token and save it somewhere you will remember, you will need it later!



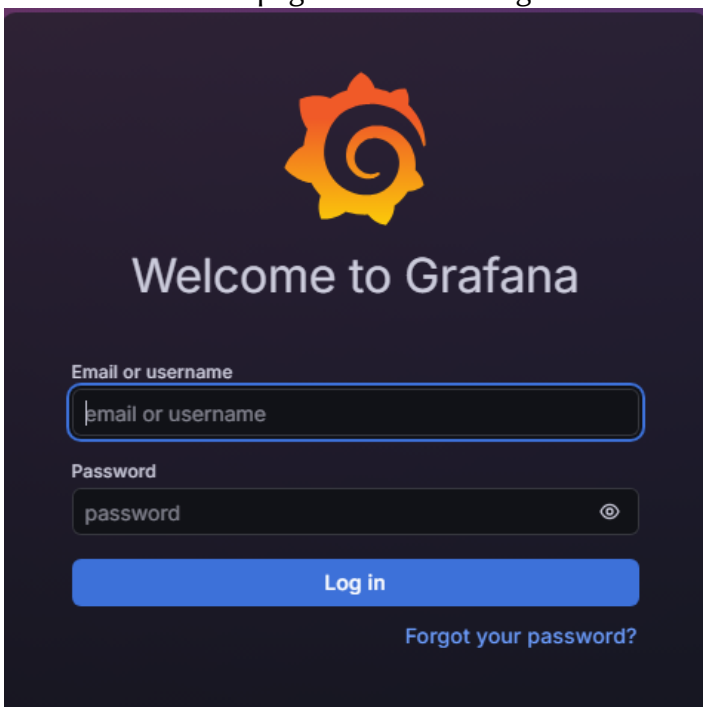
Click "Configure Later" once you have saved your token.

You are now done setting up InfluxDB. Feel free to close this tab.

[6] User data is stored locally, and not in the cloud or the Guidodyno Data Digestion repository. There is no way to recover a lost username/password, so make it simple. It is only accessible when signed into your Windows user account.

2.7 Configuring Grafana

With the runMe.bat script running from step 3.2.1, open a new browser tab and go to the Grafana local webserver at localhost:3000. You should see a page like in the image below.

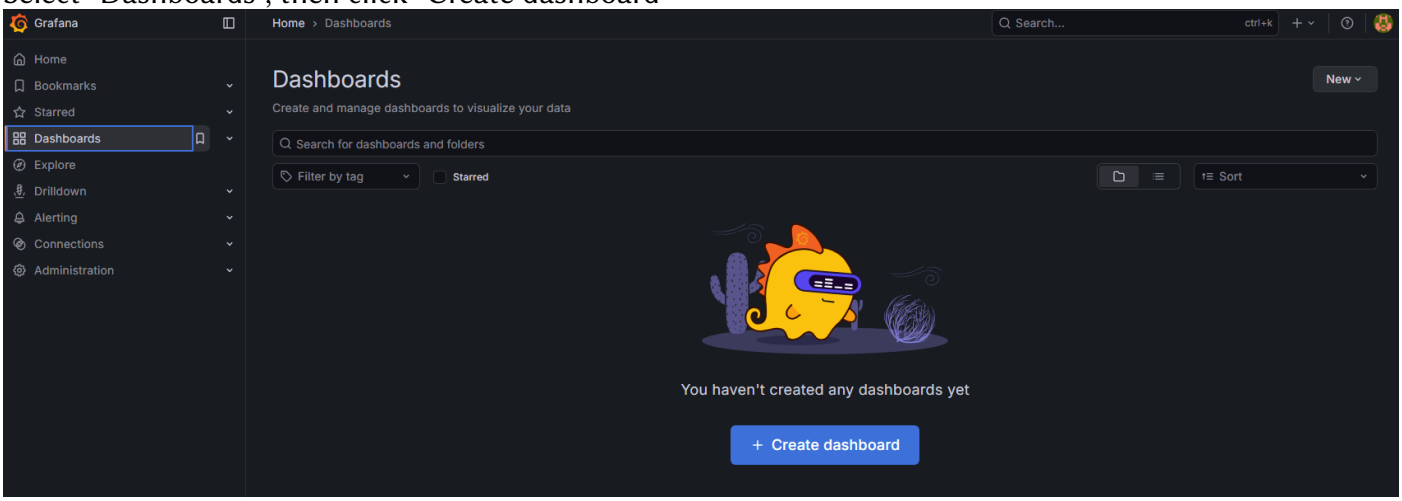


Error: Localhost Refused to Connect

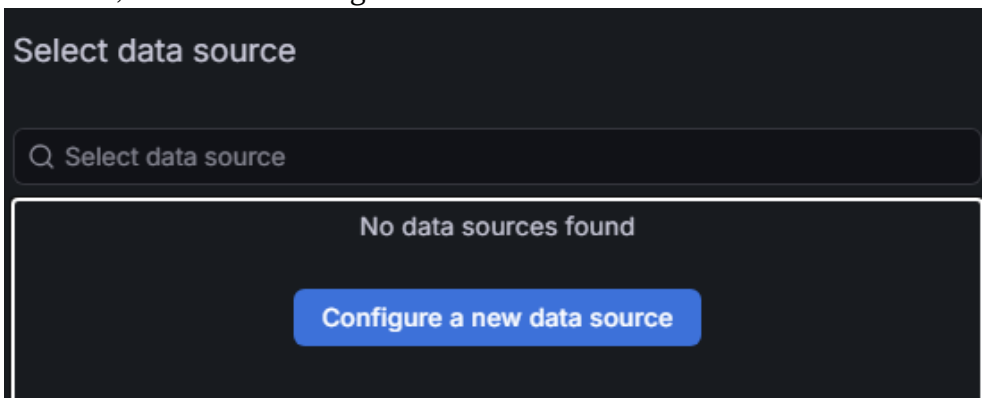
Ensure the runMe.bat program is running (as administrator), and that you have fetched the latest version of the repository (With Git LFS enabled) from Github Desktop.

The default username is "admin", and the default password is "admin". It will then prompt you to change the password. Like InfluxDB, this is only accessible within your Windows user account, so retype "admin" as your new password.

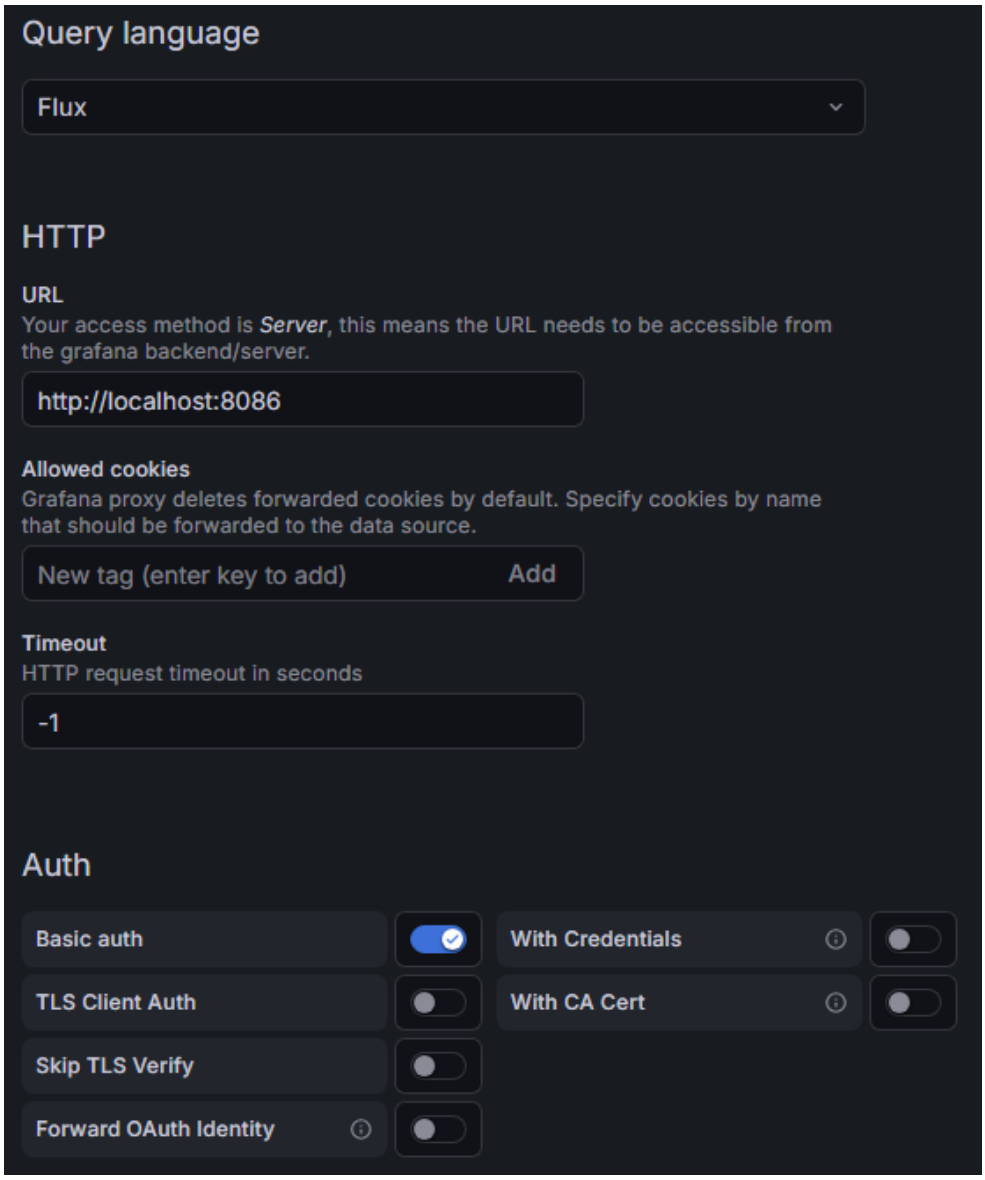
Select "Dashboards", then click "Create dashboard"



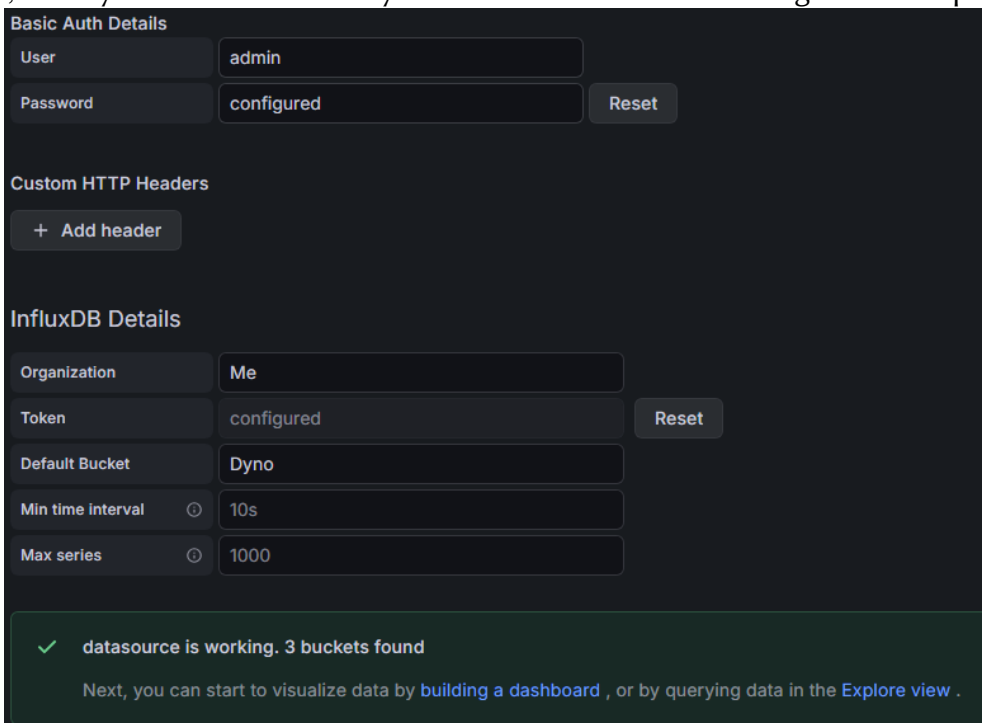
Click "Add visualization", then click "Configure a new data source"



In the next menu, search for and click on "influxdb".
Under "Query language", select "Flux".
Under "URL", enter "http://localhost:8086".
Leave "Allowed cookies" empty.
Under "Timeout", enter "-1".
Under "Auth", enable "Basic auth". Leave all other sliders off.



Right of "User", enter your InfluxDB username.
Right of "Password", leave blank. This will autopopulate in a later step.
Right of "Organization", enter "Me".
Right of "Token", enter your InfluxDB API key saved from 3.4 InfluxDB configuration steps.



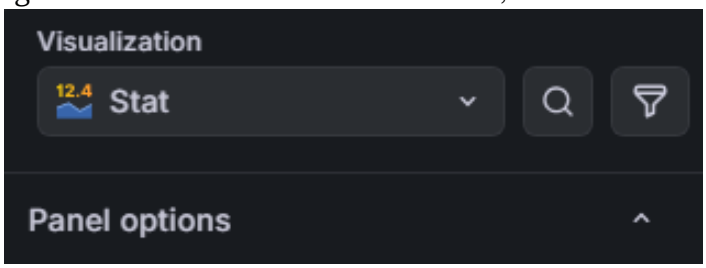
Click "Save & test". Grafana should respond that it found buckets attached to InfluxDB. [7]

Temporary UI Credential Bodge

At the moment, user credentials need to be hardcoded into the UI python program. This will be fixed in a later update with local user credential files, but for now:
In file explorer, go to the software repository.
Edit userTerminal.py with your favorite text editor.
fill in these lines with your values:
self.INFLUX_URL = "http:localhost:8086"
self.TOKEN = [PUT YOUR INFLUXDB TOKEN HERE]
self.ORG = "Me"
self.BUCKET = "Dyno"
You will have to re-edit the token field everytime you fetch the latest version of software from github until this is fixed.

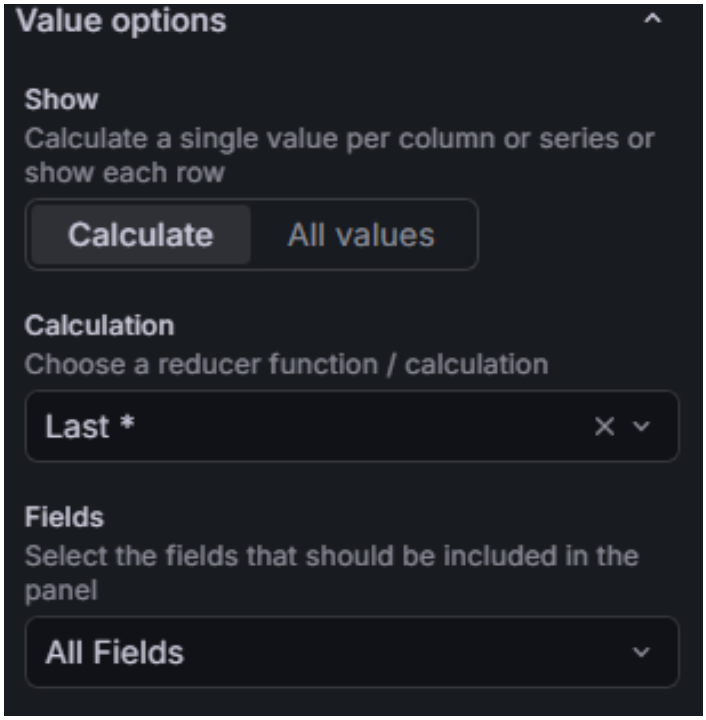
Select "Dashboards" from the left menu, click "Create dashboard", click "Add visualization", then click "influxdb" from the "Select data source" menu. This will open a new visualization pulling data from Influx.

To test our system, in the right hand menu under "Visualization", select "Stat".

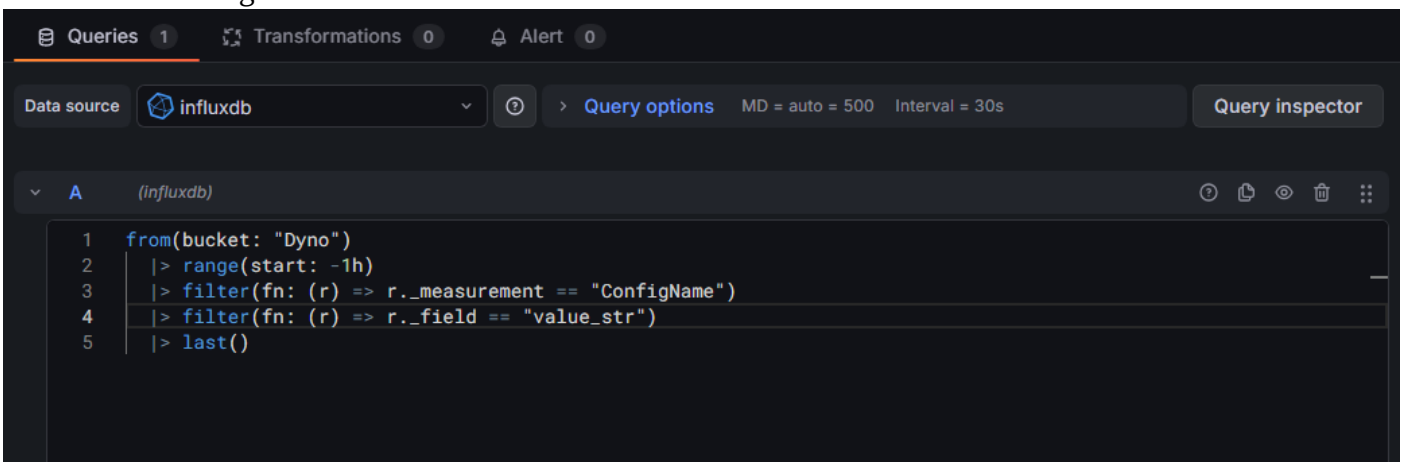


[7] If you lost your InfluxDB API token, see [TODO:insert troubleshooting section later]

Also in the right side menu, under "Fields", select "All Fields"

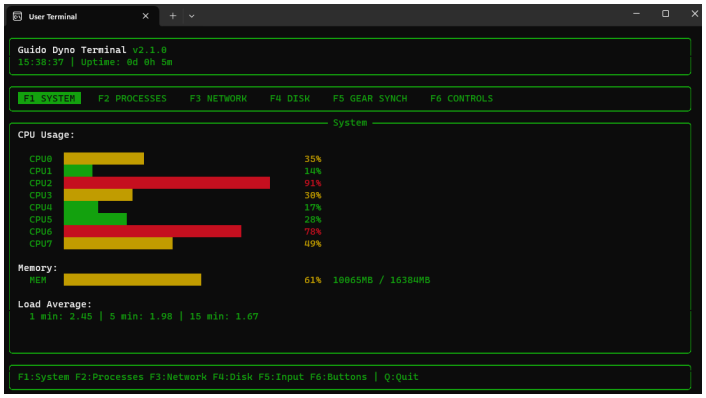


Paste the following code into the text field in the center bottom of Grafana.

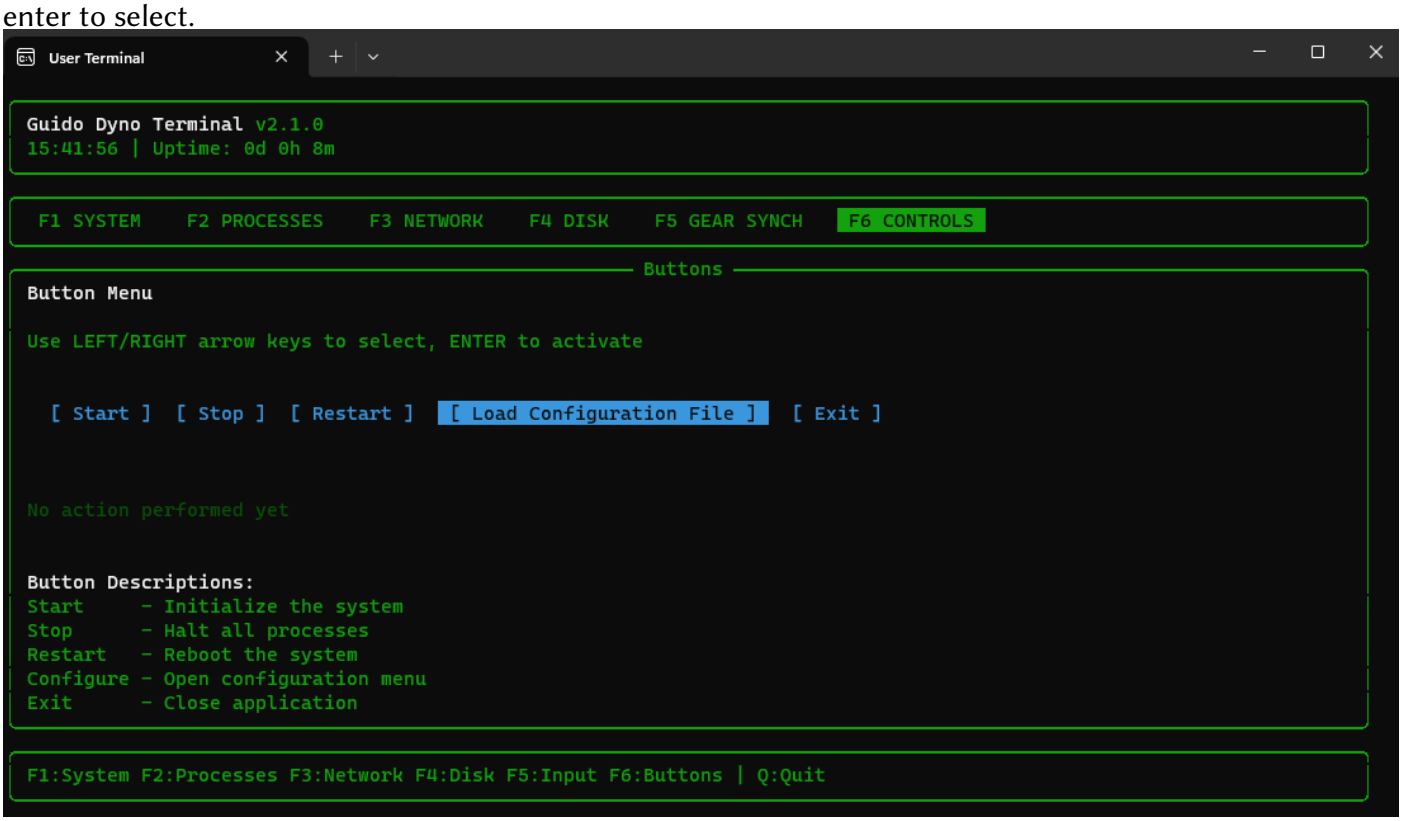



```
from(bucket: "Dyno")
  |> range(start: -1h)
  |> filter(fn: (r) => r._measurement == "ConfigName")
  |> filter(fn: (r) => r._field == "value_str")
  |> last()
```

Open the Dyno UI. This program started with the runMe.bat program, and is running in a CMD window. The splash screen looks like this.

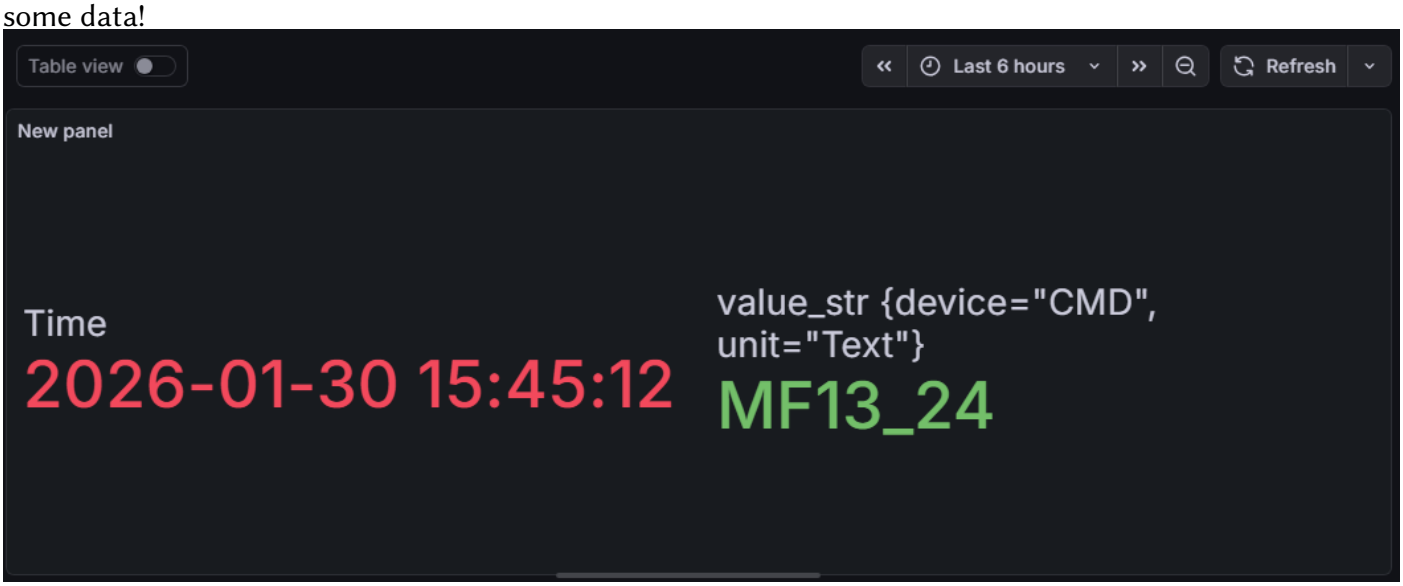


Click F6 to move to the Controls tab, then use the arrow keys to move to "Load Configuration File". Press enter to select.



Navigate to the config folder in the repository folder and open "example.json". This config folder is where Dyno configuration files are stored.

Back on the Grafana dashboard, click refresh in the top right[8]. Your stat visualization should now display some data!



If that worked, your system is configured and ready to collect data from the Guidodyno!

[8]The refresh button in the Grafana window, not the browser refresh button

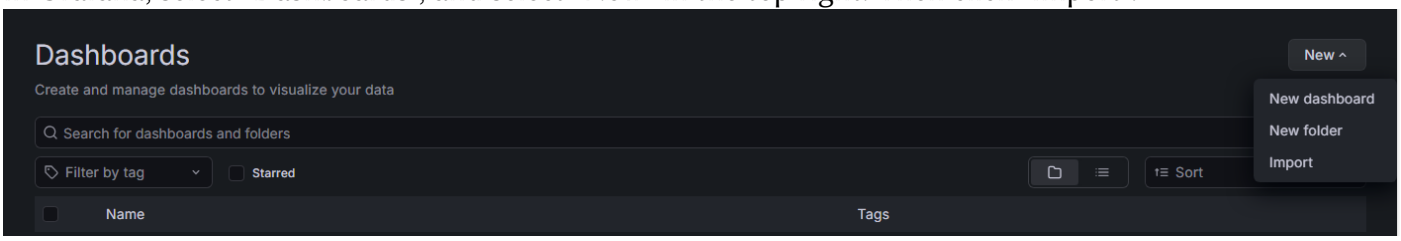
3 Use

3.1 Starting the System

The runMe.bat script, located in the top level repository folder, starts all necessary software to run the Guidodyno. Make sure the dyno is connected to your computer over Ethernet if you are collecting data before running this script. Always run the script as an administrator.

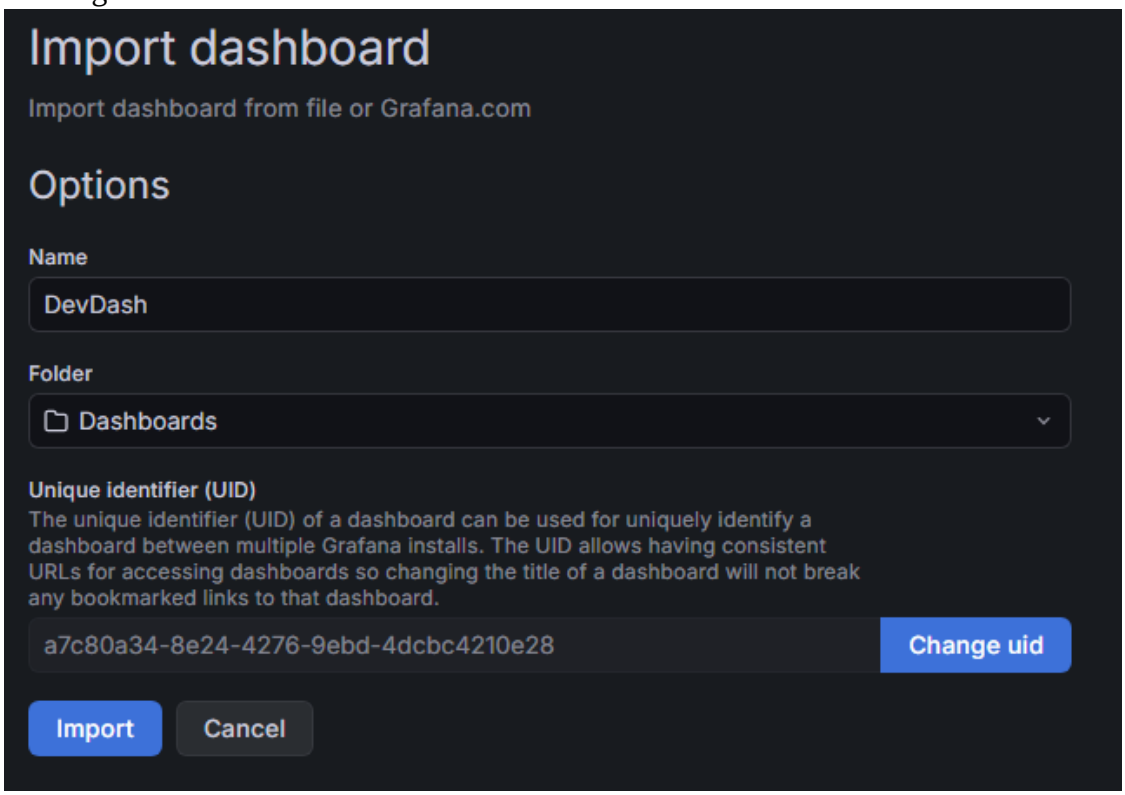
3.2 Importing Grafana Dashboards

In Grafana, select "Dashboards", and select "New" in the top right. Then click "Import".



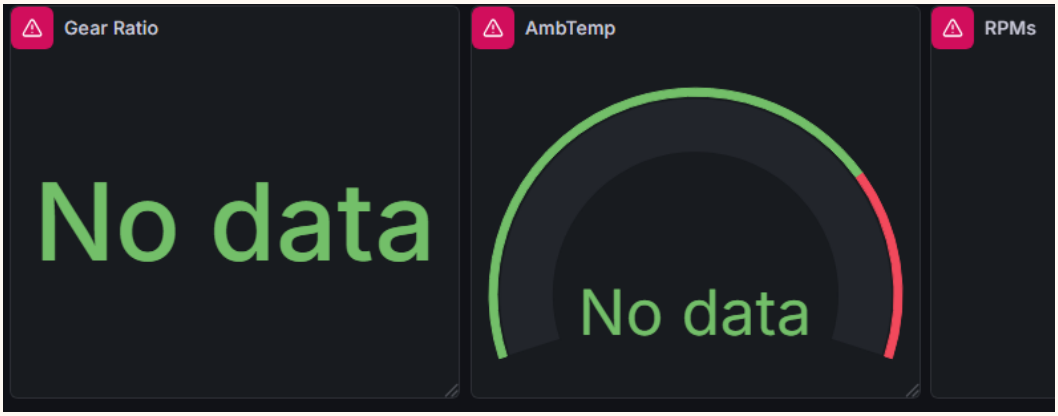
Click "Upload dashboard JSON file", and navigate to the Dyno software repository. There is a top level folder called "dashboards" that contains dashboards made by other Dyno users you can use. "DevDashboard1.json" is one that is used for testing the system.

Select the dashboard you want to import, and give it a name. Leave the Folder and UID fields alone unless you need to change them.



Click "Import". The dashboard will now load.

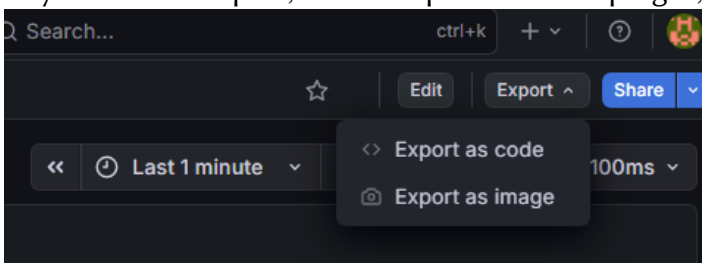
Bucket Name Mismatch



Some Dyno users use different bucket names for various reasons. A likely cause of visualization errors (red triangle in top left of visuals) is the wrong bucket name. To fix this, click the three dots in the top right of a visualization, click "Edit", then change the bucket field to your bucket name. This needs to be done for each visualization, but only once after importing a dashboard. It is now saved on your computer, and can be exported with your bucket names.

3.3 Exporting Grafana Dashboards

With the Grafana dashboard you want to export, select "Export" in the top right, and click "Export as code".



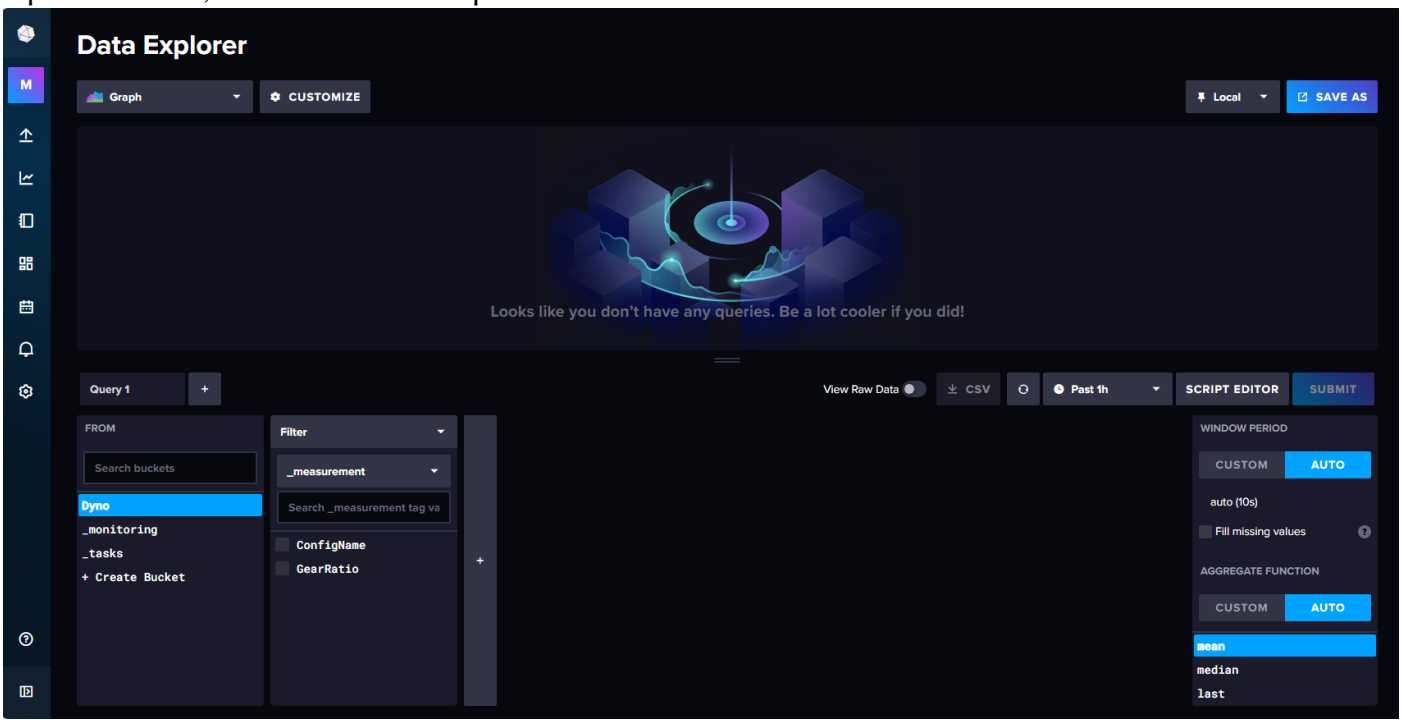
Click "Download file" in the bottom left of the download menu.

There is an shared dashboards folder in the software repository you can add your dashboard to if you would like to share with others.[9]

[9] Folder is named "dashboards"

3.4 Exporting Data to CSV

Open InfluxDB, and select "Data Explorer" form the left side menu.



Using the check boxes, select the data you want to export. Once selected, select how far you want to lookback, and click the blue "Submit" button. Once complete, click the CSV download button, next to the "View Raw Data" slider. This will download your selected data with time basis information.

3.5 View Previous Data with Grafana

TODO